

| id | title                                 | description                                                                                                             | year_of_study | create           |
|----|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------|------------------|
| 1  | Introduction to Programming           | Core programming concepts using Python, covering variables, control flow, functions, and basic data structures.         | 1             | 2026-05-13 20:06 |
| 2  | Mathematics for Computing             | Essential mathematics including logic, sets, discrete structures, and introductory calculus for computer scientists.    | 1             | 2026-05-13 20:06 |
| 3  | Data Structures & Algorithms          | Fundamental algorithms, complexity analysis, sorting, searching, trees, and graphs.                                     | 2             | 2026-05-13 20:06 |
| 4  | Software Engineering                  | Agile methodologies, design patterns, version control, testing, and professional software development practices.        | 2             | 2026-05-13 20:06 |
| 5  | Final Year Project                    | Independent research and development project supervised by a member of academic staff.                                  | 3             | 2026-05-13 20:06 |
| 6  | Principles of Management              | Introduction to organisational behaviour, leadership styles, and management theory.                                     | 1             | 2026-05-13 20:06 |
| 7  | Marketing Fundamentals                | Core marketing concepts including the marketing mix, consumer behaviour, and digital marketing.                         | 1             | 2026-05-13 20:06 |
| 8  | Financial Accounting                  | Fundamentals of financial reporting, balance sheets, income statements, and basic ratio analysis.                       | 2             | 2026-05-13 20:06 |
| 9  | Machine Learning                      | Supervised and unsupervised learning, neural networks, model evaluation, and scikit-learn/TensorFlow practicals.        | 1             | 2026-05-13 20:06 |
| 10 | Big Data Technologies                 | Hadoop, Spark, and cloud-based data pipelines for processing and analysing large-scale datasets.                        | 1             | 2026-05-13 20:06 |
| 11 | Statistical Methods                   | Probability, hypothesis testing, regression analysis, and Bayesian inference applied to real-world datasets.            | 1             | 2026-05-13 20:06 |
| 12 | Network Security Fundamentals         | TCP/IP security, firewalls, intrusion detection, VPNs, and cryptography basics.                                         | 1             | 2026-05-13 20:06 |
| 13 | Ethical Hacking & Penetration Testing | Practical penetration testing methodologies, tools (Metasploit, Burp Suite), and responsible disclosure.                | 2             | 2026-05-13 20:06 |
| 14 | Web Development                       | Frontend and backend web development using HTML, CSS, JavaScript, PHP, APIs, and responsive design principles.          | 1             | 2026-05-13 20:18 |
| 15 | Object-Oriented Programming           | Core object-oriented concepts including classes, inheritance, abstraction, encapsulation, and reusable software design. | 1             | 2026-05-13 20:18 |
| 16 | Database Systems                      | Relational database design, SQL, normalization, indexing, and transaction management for robust applications.           | 2             | 2026-05-13 20:18 |
| 17 | Requirements Engineering              | Methods for eliciting, analysing, documenting, and validating software and system requirements.                         | 2             | 2026-05-13 20:18 |
| 18 | DevOps and Deployment                 | Continuous integration, delivery pipelines, containerisation, monitoring, and deployment automation.                    | 3             | 2026-05-13 20:18 |
| 19 | Corporate Finance                     | Capital structure, investment appraisal, financial planning, and decision-making within business organisations.         | 2             | 2026-05-13 20:18 |
| 20 | Taxation Principles                   | Introduction to personal and corporate taxation, compliance, tax calculations, and legal responsibilities.              | 2             | 2026-05-13 20:18 |
| 21 | Auditing                              | Internal and external auditing concepts, assurance, audit evidence, controls, and ethical issues.                       | 3             | 2026-05-13 20:18 |
| 22 | Digital Content Strategy              | Planning and managing digital content for websites, campaigns, email, and social platforms.                             | 1             | 2026-05-13 20:18 |
| 23 | Social Media Analytics                | Measurement of campaign performance using engagement, reach, conversion, and reporting metrics.                         | 2             | 2026-05-13 20:18 |
| 24 | Consumer Behaviour                    | How consumers think, decide, and respond to marketing messages across digital and traditional channels.                 | 2             | 2026-05-13 20:18 |
| 25 | Campaign Planning                     | Design and execution of integrated marketing campaigns with budgeting and performance evaluation.                       | 3             | 2026-05-13 20:18 |
| 26 | Engineering Mathematics               | Applied mathematics for engineers including calculus, matrices, differential equations, and numerical methods.          | 1             | 2026-05-13 20:18 |
| 27 | Statics and Dynamics                  | Forces, motion, equilibrium, and engineering problem-solving in mechanical systems.                                     | 1             | 2026-05-13 20:18 |
| 28 | Thermodynamics                        | Energy transfer, heat, work, and thermodynamic systems in engineering applications.                                     | 2             | 2026-05-13 20:18 |
| 29 | Materials and Manufacturing           | Engineering materials, properties, production processes, and manufacturing systems.                                     | 2             | 2026-05-13 20:18 |
| 30 | Machine Design                        | Mechanical component design, stress analysis, fatigue, and design for safety and reliability.                           | 3             | 2026-05-13 20:18 |
| 31 | Deep Learning                         | Neural networks, backpropagation, convolutional networks, transformers, and practical model development.                | 1             | 2026-05-13 20:18 |
| 32 | Natural Language Processing           | Text preprocessing, language models, classification, information extraction, and NLP applications.                      | 1             | 2026-05-13 20:18 |
| 33 | Computer Vision                       | Image processing, feature extraction, object detection, segmentation, and visual recognition systems.                   | 1             | 2026-05-13 20:18 |
| 34 | AI Ethics and Governance              | Bias, fairness, transparency, accountability, privacy, and regulation in AI systems.                                    | 1             | 2026-05-13 20:18 |
| 35 | Project Planning and Control          | Project scheduling, milestones, work breakdown structures, and performance tracking methods.                            | 1             | 2026-05-13 20:18 |
| 36 | Risk and Quality Management           | Identification of project risks, mitigation planning, quality assurance, and governance practices.                      | 1             | 2026-05-13 20:18 |
| 37 | Agile Project Delivery                | Scrum, Kanban, stakeholder communication, sprint planning, and iterative delivery methods.                              | 1             | 2026-05-13 20:18 |
| 38 | Procurement and Contracts             | Supplier management, procurement processes, contract administration, and legal considerations.                          | 1             | 2026-05-13 20:18 |
| 39 | International Commercial Contracts    | Legal principles governing international business agreements and contract enforcement.                                  | 1             | 2026-05-13 20:18 |
| 40 | Trade Law and Regulation              | Regulatory frameworks affecting international trade, market access, and compliance.                                     | 1             | 2026-05-13 20:18 |
| 41 | Corporate Governance                  | Board structures, accountability, ethics, compliance, and governance frameworks in organisations.                       | 1             | 2026-05-13 20:18 |
| 42 | International Dispute Resolution      | Arbitration, mediation, litigation strategy, and enforcement across jurisdictions.                                      | 1             | 2026-05-13 20:18 |
| 43 | Distributed Systems                   | Concepts of scalability, consistency, fault tolerance, messaging, and distributed application design.                   | 1             | 2026-05-13 20:18 |
| 44 | Cloud Infrastructure                  | Compute, storage, networking, cloud service models, and architecture design in public cloud platforms.                  | 1             | 2026-05-13 20:18 |
| 45 | Containers and Kubernetes             | Containerization, orchestration, scaling, service discovery, and deployment using Kubernetes.                           | 1             | 2026-05-13 20:18 |
| 46 | Cloud Security                        | Identity, access control, encryption, compliance, threat modelling, and cloud-native security practices.                | 1             | 2026-05-13 20:18 |